Multi-Agent Academic Helpdesk

Evaluation Report

Google ADK · Ollama · ChromaDB RAG

Team members

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Repository: https://github.com/Arinpatel18/Multi-Agent-Academic-Helpdesk

Raw traces and full tables: `eval/results/` (`RESULTS.md`, `per_question.csv`, `run*.jsonl`)

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1. Summary

We built an IIT Jodhpur academic helpdesk with Google ADK, Ollama models (through ADK's LiteLLM connector), and a ChromaDB RAG pipeline over 27 documents. A single-agent baseline was extended to a multi-agent system: an orchestrator transfers each question to a **policy**, **schedule** or **email** agent. We evaluated it on 30 questions, with 3 repeated runs per configuration.

Resource constraints: the assignment suggests 8B models on a 16 GB machine. Our only available machine was an **Apple M1 Mac with 8 GB RAM**, where an 8B model could not run at a usable speed. We therefore used a **4B-parameter model (Qwen3-4B-Instruct)** as the main model, and a 3B model (Llama 3.2) for the model-replacement comparison (§4.1). The code supports 8B models unchanged.

Key results:

Configuration	Accuracy	Correct & grounded	Correct refusals (out-of-scope)	False refusals	Routing
Multi-agent, qwen3-4b-instruct (3 runs)	**88%** (86% after manual review)	**83%**	**100%**	**8%**	**100%**
Single-agent, qwen3-4b-instruct (1 run)	50%	50%	100%	33%	–
Multi-agent, llama3.2:3b (3 runs)	39%	38%	100%	36%	64%

- With the **same model**, the **multi-agent design raises accuracy from 50% to 88%**. Smaller, focused instructions and tool sets make a 4B model choose the right tool far more reliably.
- **Out-of-scope questions were always refused (100%)** in every configuration, and every refusal ended up in the human-review log.
- **Replacing the LLM matters as much as the architecture.** llama3.2:3b over-refuses and often emits tool calls as plain text, which drops accuracy to 39%.
- **qwen3-4b-instruct never changed a correct/incorrect outcome across 3 runs.** The wording changed on 5 questions, and one of those changes (P02, run 1) was substantive.

2. System

Part	Implementation
Framework	Google ADK 2.13: `LlmAgent`, `sub_agents` + `transfer_to_agent`, `Runner`, `InMemorySessionService`
LLM	Ollama through `LiteLlm("ollama_chat/<model>")`, temperature 0.2, context 8192
Embeddings	`nomic-embed-text` (Ollama), with `search_document:` / `search_query:` prefixes
Vector store	ChromaDB (cosine). Page-aware chunks of about 1,200 characters with 200 overlap: 2,226 chunks from 24 documents
Retrieval	Hybrid: dense top-20 + BM25 top-20, fused with Reciprocal Rank Fusion; top 6 returned with `file, page` citations
Agents	`helpdesk_orchestrator` → `policy_agent` (rules, curriculum, calendar), `schedule_agent` (slots, offerings, counts), `email_agent` (contacts, drafts)
Tools	`search_academic_documents`, `lookup_course`, `list_offered_courses`, `get_slot_timing`, `get_instructor_contact`, `draft_email` (saved to disk, never sent), `log_unanswered_question`
Unanswered log	`logs/unanswered_questions.jsonl`. The agents call the tool, and the runner adds a safety net that logs any refusal the model did not log itself

The single-agent baseline (`--mode single`) is one `LlmAgent` with all seven tools and a combined instruction.

3. Data

27 documents (manifest and source URLs: `data/README.md`):

- **24 real IITJ documents**, downloaded from the public Office of Academics pages:
  - UG/PG academic regulations (2019 and 2022)
  - a 2020 amendment notification
  - circulars on the AIIMS joint programmes and M.S. by Research
  - academic calendars for AY 2025-26, AY 2026-27 and Summer 2026
  - 11 curricula and course booklets
  - the first-year slot-system spreadsheet
  - three web pages: UG registration guidelines, the programme list, and specialisations/minors
- **3 structured CSVs** for the current semester's course offerings, slot timetable and faculty directory. The real versions are only on the IITJ intranet, so these were reconstructed in the official format: real course titles and HSS course codes, with **dummy instructor names and emails**.

Personal data removed:

- Faculty names and emails in the slot spreadsheet were replaced with dummies.
- A roll-number list was dropped.
- Duplicates and scanned PDFs with no text layer were also removed.

4. Evaluation method

4.1 Resource constraints: 4B model on an M1 Mac

The assignment targets 8B models on a 16 GB machine. Due to resource constraints, all experiments were run on an **Apple M1 Mac with 8 GB of unified memory**. We tried `qwen3:8b` first: its 5.2 GB of weights plus the context cache did not fit alongside the operating system, so the machine swapped heavily and generated **0.02 tokens/s** (one 30-token answer took 24 minutes).

We therefore used the **4B-parameter** `qwen3:4b-instruct-2507` **(4-bit, about 3.8 GB)** as the main model. It runs fully on the M1 GPU at 10–40 s per question. The model-replacement comparison uses `llama3.2:3b`. The code defaults to `qwen3:8b`, and `./run_evaluation.sh` reproduces the full study with `qwen3:8b` and `llama3.1:8b` on a 16 GB machine (§6).

We also had to choose the qwen3 build carefully. Plain `qwen3:4b` is a thinking-only build: it ignored `think=false` and took about 200 s per question. `qwen3:4b-instruct-2507` needs about 20–30 s.

4.2 Question set (`eval/questions.json`)

30 questions with ground-truth answers verified against the documents, plus the expected source and the expected agent for each:

Category	n	Examples
Policy (RAG)	13	attendance for the major exam, open-elective credits, make-up deadline, grade points, minor eligibility, exam dates, M.Tech max duration, course prerequisites
Schedule (CSV)	7	slot of Autonomous Systems, number of 400-level HSS electives, instructor of CSL7570, class timings, a non-existent course
Email	4	leave email for CSL7570, make-up exam request, sick leave with name and roll number, instructor email ID

Category	n	Examples
Out of scope	6	dining, reaching campus, IPL, Wi-Fi password, JEE closing rank, mess fee

All five example questions from the assignment are included (P01, P02, S01, S02, E01).

4.3 Metrics (`eval/score.py`)

- **Accuracy**: an in-scope answer is correct if it matches every required fact pattern (regex). Text changes across runs were also **reviewed manually** (§5.5).
- **Groundedness**: the answer cites an expected source document *and* that document was actually returned by a tool call in the same run. This catches invented citations.
- **Correct & grounded**: both of the above.
- **Correct refusal**: an out-of-scope question was refused. **Logged by agent**: the agent itself called `log_unanswered_question`.
- **False refusal**: an in-scope question was refused or left unanswered.
- **Routing**: the final answer came from an expected agent.
- **Changes across runs**: the correct/refused/answering-agent outcome flipped, or the answer text similarity fell below 0.9.

5. Results

5.1 Overall

Accuracy (in-scope questions, all runs pooled):

Configuration	Runs	Overall	Policy	Schedule	Email	Grounded	Correct & grounded
qwen3-4b-instruct, multi	3	88% (63/72)	85%	86%	100%	83%	83%
qwen3-4b-instruct, single	1	50% (12/24)	46%	71%	25%	67%	50%
llama3.2:3b, multi	3	39% (28/72)	44%	43%	17%	50%	38%

Refusals, routing and speed:

Configuration	Out-of-scope refused	Out-of-scope logged by agent	False refusals	Routing	Mean latency
qwen3-4b-instruct, multi	100% (18/18)	0%	8%	100%	25.9 s
qwen3-4b-instruct, single	100% (6/6)	0%	33%	–	21.5 s
llama3.2:3b, multi	100% (18/18)	100%	36%	64%	11.6 s

qwen3-4b-instruct multi-agent answered the five assignment example questions correctly and with citations in all 3 runs, except P02 (correct in 2 of 3):

- **75% attendance:** students below it may still appear (01_Academic_Regulations p.7).
- **10 open-elective credits:** in 2 of 3 runs (see §5.5).
- **Autonomous Systems:** slot J.
- **400-level HSS electives:** 5.
- **CSL7570 leave email:** a draft to the instructor's address, saved to disk.

5.2 Single agent vs. multi-agent (same model)

The multi-agent version is **38 points more accurate** and has **4× fewer false refusals**. The single agent's failures were all tool-selection problems:

- **No tool call at all (6 policy questions: P02, P05, P08, P09, P10, P13).** It refused immediately without searching the documents, even though the multi-agent policy agent answered all of them except P13.
- **Bypassing the email tool (E01, E02).** It wrote the email itself instead of calling `draft_email`, so the draft has no instructor address and was not saved.
- **Wrong tool (E04, S07).**
  - **E04:** it called `lookup_course` instead of `get_instructor_contact` and replied that the email ID "is not available".
  - **S07:** it searched the curricula instead of the course-offering list.

With 7 tools and one long instruction, a 4B model loses track of which tool to use. With 2–5 tools and a short, role-specific instruction per agent, it rarely does.

5.3 Replacing the LLM (qwen3-4b-instruct → llama3.2:3b)

llama3.2:3b's errors break down into three groups:

- **Over-refusal at the orchestrator (21 of 72 in-scope answers).** It classified grading, exam dates and programme questions as out of scope.
- **Tool calls emitted as text (9 answers).** For example, E01 returned the literal string `{"type":"function","function":"email_agent",...}` instead of a real transfer. As a result, no email draft was ever saved.
- **Empty answers (11).** A specialist called the right tool and then replied with only `Source: course_offerings…csv` (S05, S06, S07).

Its 100% out-of-scope refusal rate is partly a side effect of refusing too much. It should be read together with its 36% false-refusal rate. It was, however, the only model that called `log_unanswered_question` itself.

5.4 Groundedness and refusals

- **Citations were real.** For qwen multi, every answer that cited the expected source cited a document the tools had actually returned in that run. No citation was invented.
- **One unsupported inference (P13).** For "In which year did the B.Tech AI&DS start?", retrieval returned only spreadsheet chunks. The model "inferred" 2023 from roll-number prefixes instead of refusing. The correct answer is 2020, from doc 31. This was the only answer that went beyond its sources.
- **Logging relies on the safety net for qwen.** The qwen orchestrator never called the logging tool: it printed the refusal sentence ("…I have logged your question…") without making the call. All 24 qwen refusals still reached the log through the runner's safety net. **Logging must not depend on the model's cooperation.**
- **Specialist refusal without searching (P11, S04).** In all 3 runs, the specialist refused without calling any tool and printed a fake `log_unanswered_question(...)` as text. Both questions are answerable.

5.5 Questions whose answers changed across runs

Config	Outcome changed (correct/refused/agent)	Text changed only
qwen3-4b-instruct, multi	none	P01, **P02**, P04, P13, S02
llama3.2:3b, multi	**E01, E04, P03, P11**	P06, P10
qwen3-4b-instruct, single	(1 run, not measured)	–

- **P02 (qwen) changed substantively.** In run 1 the model read the wrong table row and answered with the 10 open-elective credits that count toward a *minor*. Runs 2–3 correctly answered "10 OE credits for a regular B.Tech". The regex scorer accepted all three runs because each contains "10". After manual review, run 1 is wrong, so **manual accuracy is 62/72 = 86%**.

- **P01, P04, P13 and S02 (qwen)** were rephrasings with the same facts.
- **llama3.2:3b:**
  - **E01:** the transfer was emitted as text in some runs.
  - **E04:** answered in 2 of 3 runs.
  - **P03:** correct in 1 of 3 runs.
  - **P11:** routed to the schedule agent in one run and to the policy agent in the others.

5.6 A conflict between the documents (P06)

The **2019 regulations** (doc 01, p.16) say a student needs **100 credits by the end of the 6th semester** to opt for a minor. The **current IITJ web page** (doc 32) says **85 graded credits by the end of the 5th semester**. The agent consistently answered with the newer rule and cited doc 32. Both answers were accepted, since each is grounded. A production helpdesk should prefer the newest source and flag the conflict to the student.

5.7 Development iterations

Traces are kept in `eval/dev_iterations/`. The first trial run exposed three problems, which were fixed before the reported runs:

1. **Padded search queries.** The model added "IIT Jodhpur academic regulations" to its queries. Those words appear in every chunk, which drowned the useful terms and caused misses on P03 and P05. Fix: strip these filler phrases from queries, and instruct the agent to use short keyword queries and search a second time before giving up.
2. **Make-up exam rule misrouted.** P04, a question about the make-up exam *rule*, was sent to the email agent. Fix: route a question to the email agent only when the user explicitly asks for an email.
3. **Fuzzy matching too loose.** A unit test showed that "Quantum Machine Learning" resolved to "Machine Learning". Fix: stricter title matching.

6. Running with 8B models

Because of the resource constraints in §4.1, the 8B configurations were not evaluated. The system is not tied to the 4B model: the default model is `qwen3:8b`, and on a 16 GB machine one command reproduces the full study with the assignment's suggested models:

```
./setup.sh            # dependencies, qwen3:8b + llama3.1:8b + embeddings, index
./run_evaluation.sh   # qwen3:8b multi x3, llama3.1:8b multi x3, single x1, scoring
```

We expect 8B models to reduce the two main failure types we saw: skipped tool calls (qwen 4B) and over-refusal or tool calls printed as text (llama 3B). Both are known weaknesses of small models at tool calling.

7. Limitations and future work

- **Scoring:** the regex scorer is strict on facts but can accept a right number given for the wrong reason (P02). Manual review or an LLM judge is needed for a final number.
- **Reconstructed CSVs:** the current-semester CSVs are reconstructions, so schedule results show tool use, not real timetable accuracy.
- **Tool skipping (P11, S04, single-agent P05):** could be fixed by requiring a tool call on a specialist's first turn (ADK `tool_config` mode `ANY`), or by having the runner call `search_academic_documents` directly before the LLM answers.
- **Logging:** make logging fully deterministic. Any route to the refusal path writes to the log in code, which is already true through the safety net.
- **Retrieval:** add a cross-encoder reranker and table-aware PDF parsing. Several facts sit in flattened tables (credit structure, grade points), and adding the year and title as metadata would favour the newest source when documents conflict.
- **Model choice:** a stronger tool-calling model (8B) should reduce both the over-refusal seen with llama3.2:3b and qwen's skipped tool calls. §6 gives the commands to run this comparison on a 16 GB machine.